Problem 1

$x = \pi$

$y_1 = 22/7$ $\qquad |x - y_1| = 1.3 \times 10^{-3}$ ← absolute error

$\dfrac{|x - y_1|}{|x|} = 4.0 \times 10^{-4}$ ← relative error

$y_2 = 355/113$ $\qquad |x - y_2| = 2.7 \times 10^{-7}$

$\dfrac{|x - y_2|}{|x|} = 8.5 \times 10^{-8}$

both the absolute and the relative errors are smaller for y_2 and therefore it is a better approximation to x.

Problem 2

1. $p(x) = x^{32} = x^{2^5}$

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for i = 1 to 5
    x ← x . x
end for
p = x
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$i = 1 \quad x = x^2$

$i = 2 \quad x = x^2 \cdot x^2 = x^4$

$i = 3 \quad x = x^4 x^4 = x^8 = x^{2^3}$

$\vdots$

2. $p(x) = (x-1)^5 \left[3 + 7(x-1)^4 \right]$

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z ← x - 1
p ← z^5 [ 3 + 7 z^4 ]
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3. $p(x) = (x+2)^3[6 + (x+2)^4[9 + (x+2)^8[3 - (x+2)^{16}]]]$

$$\left[\begin{array}{l} z \leftarrow x+2 \\ p \leftarrow z^3[6 + z^4[9 + z^8(3 - z^{16})]]. \end{array}\right.$$

4. $p(x) = [[[x^{90} - 5]x^{20} + 10]x^{10} - 3]x^7$

$$[p \leftarrow (((x^{90} - 5)x^{20} + 10)x^{10} - 3)x^7$$

<u>Problem 3</u>

$$\begin{aligned} y &= 11 + 9e^x + 7e^{2x} + 5e^{3x} \\ &= 11 + e^x(9 + e^x(7 + 5e^x)) \end{aligned}$$

$$\left[\begin{array}{l} e \leftarrow e^x \\ y \leftarrow 11 + e(9 + e(7 + 5e)) \end{array}\right.$$

<u>Problem 4</u>

1. This is an alternating series for π:

$$\pi = 4\left(1 - 1/3 + 1/5 - 1/7 + \dots\right)$$

$$= 4 \lim_{N \to +\infty} \underbrace{\sum_{n=1}^{N} \left((-1)^{n+1} \frac{1}{2n-1} \right)}_{S_N \leftarrow \text{partial sum}} = a_n$$

We want π to be rounded to 3.14. According to our definition, 2 decimal digits corresponds to $t-1 = 2$ significant digits and therefore, in rounding, $\eta = \frac{1}{2} 10^{1-t} = \frac{1}{2} 10^{-2}$.

From the Alternating Series Theorem, the error is

$$|S - S_N| \leq |a_{N+1}| = \frac{4}{2(N+1)-1} = \frac{4}{2N+1} < \eta$$

$$\Rightarrow \frac{4}{2N+1} < \frac{1}{2} 10^{-2}$$

$$\Rightarrow 2N+1 > 8\, 10^{2} = 800$$

$$\Rightarrow 2N > 799$$

$$\Rightarrow N > 399.5 \Rightarrow N = 400$$

So, 400 terms are needed.

2. The same reasoning as in 1. applies to the given series:

$$-1 + 2^{-4} - 3^{-4} + 4^{-4} - \dots = \lim_{N \to +\infty} \underbrace{\sum_{n=1}^{N} (-1)^n (n)^{-4}}_{S_N}$$

$$|a_{N+1}| = (N+1)^{-4} < \eta$$

For 4 significant digits in chopping: $\eta = 10^{-4}$
(for $(t-1)$ significant digits in chopping, $\eta = 10^{1-t}$)

$$(N+1)^{-4} < 10^{-4}$$

$$N > 10 \quad \Rightarrow \quad N = 11$$

11 terms are needed.

<u>Problem 5</u>

1. 64.01562

(i) first, we convert the integer 64:

remainder

$2 \,|\, 64 \;\cdots\; 0 = b_0$

$2 \,|\, 32 \;\cdots\; 0 = b_1$

$2 \,|\, 16 \;\cdots\; 0 = b_2$

$2 \,|\, 8 \;\cdots\; 0 = b_3$

$2 \,|\, 4 \;\cdots\; 0 = b_4$

$2 \,|\, 2 \;\cdots\; 0 = b_5$

$2 \,|\, 1 \;\cdots\; 1 = b_6$

0

or $2^6 = 64$

$$\Rightarrow \quad (64)_{10} = (\overset{b_6}{1}\,0\,0\,0\,0\,0\,\overset{b_1}{0}\,\overset{b_0}{0})_2$$

(ii) now, the decimal part 0.01562

$$\begin{aligned}
& 0.01562 \\
& \quad \times 2 \\
d_1 = {} & 0.031305 \\
& \quad \times 2 \\
d_2 = {} & 0.06261 \\
& \quad \times 2 \\
d_3 = {} & 0.12522 \\
& \quad \times 2 \\
d_4 = {} & 0.25044 \\
& \quad \times 2 \\
d_5 = {} & 0.50088 \\
& \quad \times 2 \\
d_6 = {} & 1.00176
\end{aligned}$$

$$\Rightarrow \quad (64.015625)_{10} = (1000000.000001)_2$$

$$= (1.00000000001)_2 \cdot 2^6$$

$c - 127 = 6 \Rightarrow c = (133)_{10} = (10000101)_2$

so the single precision bit string is:

$$\underset{+}{0}\ \underbrace{10000101}_{\text{exponent}}\ 000\,000\,000\,001\,0000000\,00000$$

in double precision:

$c - 1023 = 6 \rightarrow c = (1029)_{10} = (1000000101)_2$

so the floating point representation in 64 bits is

$$\underset{+}{0}\ \underbrace{1000000101}_{\text{11-bit exp}}\ \underbrace{000\,000\,000\,001\,0000000\ \ldots\ 000}_{52\text{ bits}}$$

Follow a similar procedure for 64.37109375:

$(0.37109375)_{10} = (0.010111110)_2$

so

$(64.37109375)_{10} = (1000000.010111110)_2$

$= (1.0000000101111110)_2 \times 2^6$

$c - 127 = 6 \Rightarrow c = (10000101)_2$

in single precision:

0 10000101 00000001011111 000000000

$c - 1023 = 6 \Rightarrow c = (10000000101)_2$

in double precision:

0 10000000101 0000000101111100 0 ... 0000.

2) $\underset{\downarrow \atop +}{0}\ \underbrace{01111111}_{8}\ \underbrace{0 \ldots 0}_{0}$

$(01111111)_2 = (127)_{10} \Rightarrow c = 0 \Rightarrow 2^0$

$\Rightarrow +(1.0)_2 \times 2^0 = +1.$

<u>Problem 6</u>

$$f(x) = e^x - x - 1$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots - x - 1$$

$$= \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

$$f(10^{-2}) = \frac{10^{-4}}{2} + \frac{10^{-6}}{6} + \frac{10^{-8}}{24} + \frac{10^{-10}}{120} + \ldots$$

$\approx 0.000050167087 50 \leftarrow$ approx value using Taylor series

$f(10^{-2}) \simeq 5.01671 \times 10^{-5}$

rounding

using $e^{0.01} \simeq 1.01010$,

$f(10^{-2}) = 1.01010 - 0.01000 - 1.00000$

$= 0.00010$ ← very different from the true value (approx using Taylor series).

This is a classic example of loss of significance through subtraction.

<u>Problem 7</u> Go to Matlab Central and take a look at the bubble sort implementation by Brian Moore.